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Digital and distributed project management in mechanical engineering studies – a case study

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Abstract

The acceleration of product development requires clear and systematic project management as a guideline and reference point for engineers from different backgrounds, different organizations and different cultures. Project management has therefore been part of the education of mechanical engineers for years. During the last years product development projects evolved the need for agile and lean management methods. So concepts originated from software development influence the organization of even pure mechanical engineering departments. Multidisciplinary product development projects can nowadays hardly do without scrum.

Now, the COVID-19 pandemic suddenly transformed student cohorts from local classrooms into distributed teams from one day to the other. The basics of engineering project management could neither be explained nor trained in classroom teaching. Accidental discussions did not take place. Asking was difficult and feedback loops could not be closed. Exams in presence could not be carried out and has been transformed into some kind of homework assignment. The individual student got out of focus and learning success began to slip away.

This contribution provides a case study how the sudden restrictions led to the implementation of a digital and virtual concept in teaching project management for mechanical engineering studies. It shows how classic project management is taught in a blended lecture set-up with asynchronous videos, synchronous web meetings and project based learning in a bachelor course. It further shows how agile project management is used to keep the close contact between student and teacher during student thesis. It also explains how scrum is applied to organize self-learning in a master course. The focus of this contribution is on the implementation of known methods with digital project management and communication tools. Finally, an attempt is made to compare the previous classroom teaching and the new digital teaching.

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1. Introduction

Product development needs clear and systematic project management as a guide and point of reference for engineers with different backgrounds, from different organizations and with different cultures. Mechanical engineering students learn and practice project management at various points during their studies. Nowadays, there is an ever growing focus on agile and lean management methods, because the so-called digital transformation leads to increasing interfaces to software development. Thus, multidisciplinary product development projects can hardly do without Scrum these days. Nevertheless, digital collaboration in distributed, autonomous teams is

currently not a distinct and independent learning content. However, future graduates must have these skills. A recent survey at regional machinery companies carried out by the authors clearly shows that a large proportion of respondents estimate the future share of cross-location work at 20% to 60%, see Fig. 1.

Suddenly, the COVID-19 pandemic turned student cohorts from local classrooms into distributed teams from one day to the next. As a result, the basics of project management could neither be explained nor trained in class. Comprehension questions were difficult to ask and difficult to answer, so feedback loops took a long time or could not be closed at all. Since exams could no longer be carried out in attendance, the

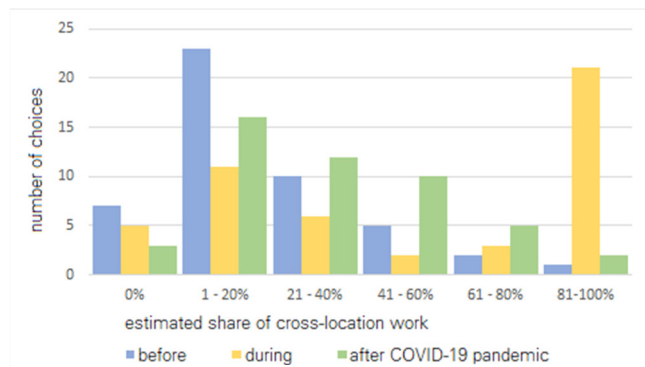


Fig. 1 Survey on working in distributed product development teams, before, during and after the COVID-19 pandemic.

learning content had to be assessed in a kind of homework. Thus, the teaching content, teaching method and exam no longer matched. The constructive aligned got lost.

These sudden restrictions led to the implementation of digital and virtual concepts in the teaching of project management for mechanical engineering studies. Classic, agile and hybrid project management were introduced in a bachelor and a master course as well as in student theses. No fundamentally new methods were invented, but known methods with digital project management and digital communication tools were applied to the courses. This contribution is intended as an exchange of experiences. The authors hope that through the exchange, good and practical ideas will be widely used and continuously improved. In this way, new teaching concepts can arise from the sudden and unexpected transition, which can be used meaningfully and profitably even after the pandemic.

2. Definitions of project management

One of three projects doesn't meet the expected project goals [1]. Project management is important to maintain competitiveness. This section gives a short overview and definition on the terms classic, agile and hybrid project management in the context of product development projects.

2.1. Classic Project Management

Within this contribution classic project management is understood as defined e.g. in [2]. The product development process is separated into distinct phases, such as initiating, planning and closing. In each phase specific process steps need to be performed and methods are used. At the end of each phase a milestone is located to assure that necessary documents have been created, such as risk assessment sheet or Gantt-chart. If the milestone is passed, the next phase can begin.

2.2. Agile Project Management

Agile project management was firstly developed in software development, but spreads out in various disciplines. In 2001 the Agile Manifesto has been formulated by Ken Schwaber, Jeff Sutherland and others and defined the four agile principles:

- Individuals and interactions over processes and tools.
- Working software over comprehensive documentation.
- Customer collaboration over contract negotiation.
- Responding to change over following a plan.

Apart from software development methods, some project management methods (e.g. scrum, Kanban boards) became also relevant for mechanical engineering and found its way into lean product development approaches, such as explained in [3]. The scrum method has recently been adapted to the use in education and therefore been labelled eduScrum [4] or Scrumban [5].

2.3. Hybrid Project Management

Engineering projects—other than pure software development projects—can often not be managed with a solely agile project management method. This is due to the complexity of interdependent process steps and dependencies. For example, a complete bill of materials needs to be released in the ERP-system to order parts and start production planning. Thus, classic and agile project management is combined depending on the specific needs. One commonly used practice is to plan the macro process according to classic project management with defined milestones and concrete work packages for the different departments. The operational work within the department is then realized by agile methods, such as Kanban boards, cp. [3], [5].

3. Case Studies

The summer semester lectures started as usual on 2nd March 2020. On Friday 13th March 2020 it has been decided to close all German universities for students. Thus, on Monday 16th March 2020 the lectures continued in online mode. Although, the authors would have preferred to introduce the new technologies step by step with a little more advance notice and more supporting staff, the COVID-19 pandemic gave the impetus to finally put into practice concepts that had been envisaged for some time now. The following sections will show the application of classic, agile and hybrid project management in three different lectures.

3.1. Classic Project Management

The lecture “Management of Product Development Projects and PDM” is held in the sixth semester of Bachelor studies in Mechanical Engineering. 40 to 60 students take part every semester. At the end of the lecture students should know the different project management tasks during all project phases. They should be able to write appropriate project proposals and do feasibility studies with risk and stakeholder management. They should perform a detailed planning with Gantt-charts, critical path analysis, team and budget planning. They should know methods to identify deviations from the planning and define suitable countermeasures. Finally, they should be able to plan a lessons-learned workshop.

In the old concept the theory was explained in frontal teaching and the lecture was accompanied by a paper-based

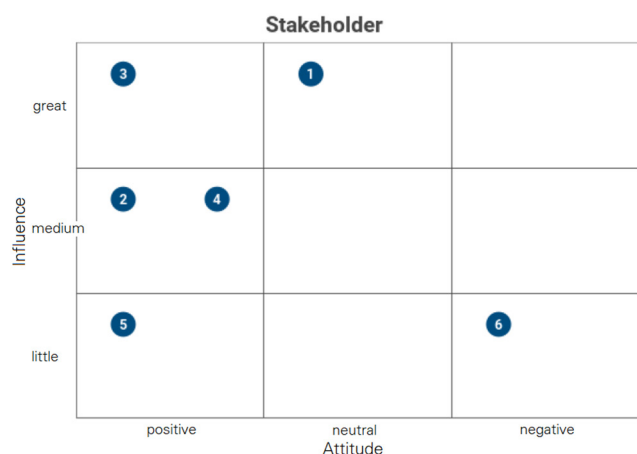


Fig. 2 Stakeholder analysis of a student team.

simulation game that was carried out during the attendance time. The intended learning outcomes have been examined through a written test. In this test, the students had to go through all project phases and carry out the project management tasks for some predefined circumstances. According to Bloom's taxonomy level 3 "application" could be reached.

The intended learning outcome was not changed for the new concept. However, the capability to use a modern, digital, collaborative project management software has been added as a new objective. Project management can be supported by several different tools, cp. [6]. For this lecture the software Projektron BCS has been chosen, because it provides a broad functionality throughout the complete project management cycle. Furthermore, it is a cloud-based software, i.e. a server is hosted at the university and students can work in their project structure from each device using a standard web browser. The access rights management guarantees, that students just work in their own project and cannot access any other project. However, the lecturers can see all the data. In order to ensure constructive alignment between what is taught in the lecture and how learning outcome is tested, the written exam has been replaced by a project work. According to Bloom's taxonomy level 6 "evaluation" should be reached.

To allow a lecture without any attendance time, the concept has been changed rapidly into some kind of project based learning concept [7]. Firstly, the project management theory was recorded and made available on video through the e-learning platform Stud.IP. Secondly, the use and operation of the software was explained in short videos and also provided via the e-learning platform. Thirdly, student-logins were created on the project management server and assigned to groups of three people each. Fourthly, a task description was drawn up for each phase. However, the task description was only published at the beginning of each individual phase. This allowed the tasks to be individualized for each team. Lastly, in an online meeting the new procedure has been explained to the students.

The students had to imagine themselves as part of the engineering team of a bike manufacturer. The management of the fictive company asked for project ideas to develop innovative smart bikes. In the first project phase the student team had two weeks to write a project proposal that fits into that scenario. The project proposal was created without any

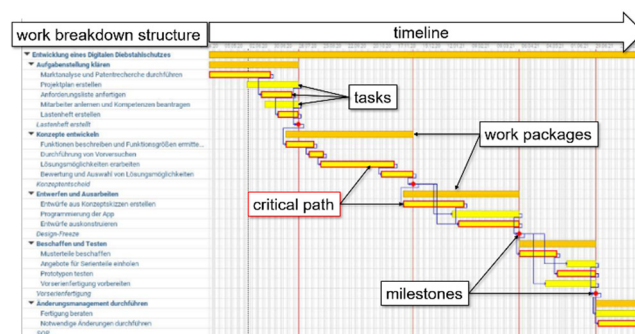


Fig. 3 Gantt-chart of a student team, red frames indicate the critical path.

project management software. Just a two-page document had to be uploaded.

The project proposal has been selected for a feasibility study, but the management changed some circumstances, e.g. lowered the budget or reduced the time to market. In the next two weeks students had to formulate objectives, perform stakeholder and risks analysis, and generate the milestone plan. In this phase the project management software was used as a tool. At the end of the phase a report has been created and uploaded. Fig. 2 shows the overview of a stakeholder analysis.

In the next phase the student teams had to generate the planning concept. Again they had to take some new management requirements into account, such as additional tasks to reduce the identified risk or shorten the development time by simultaneous engineering. At the end of this phase the project was planned with all work packages and dedicated tasks. A Gantt-chart and a critical path analysis was generated. The team planning was done and the budget planning was ready and plausible. Fig. 3 shows an example of a generated Gantt-chart from one student team.

In order to see, if the planning was good, also the realization phase has been simulated. According to the concrete project plan, the students had to react on some unforeseen events, such as team members getting ill or suppliers are not able to deliver in time. The students had to design different scenarios, evaluate them and decide for the best solution. Depending on the decision the project would not meet the time, cost or quality targets. Therefore, the students had to independently evaluate the possible solutions in the given context and decide on the best countermeasure.

In the last phase a kick-out meeting and a lessons-learned workshop had to be done. From the lessons-learned workshop clear and appropriate measures were developed to improve the next project.

The project based approach in this lecture was a clear success. Thus, it is in line with different other studies, cp. [7]. Students learned how to plan a project. Due to the immediate feedback after every phase, they could improve during the project even though no direct contact was possible. The unforeseen events that have been stated during the realization phase forced students to rethink their planning and decide for the best compromise. So even under Corona pandemic conditions we were able to raise the learning complexity from Bloom's level 3 to level 6. At the same time the grade point average could be improved by one level.

However, the effort for the lecturers during that semester was very much higher than before. That was due to the creation



Fig. 4 Virtual Reality meeting room during interactive 3D-Design review.

of videos, administration of the project management server and answering of questions. The creation of videos can be seen as an investment. Once created they can be re-used for some time. The server administration is still necessary. Although, a learning curve leads to reduced effort in the following semesters, administration needs to be planned and financed. Answering questions has been done via posts in a forum on the e-learning platform. It turned out that this procedure takes a lot of time and is prone to misunderstandings. In the following semester questions are answered in online consultations.

In the student evaluation, the lecture has an overall rating on approximately the same level as the lecture before corona pandemic. However, students complain that the time it takes to familiarize with the software and create the project plan is significantly longer than normal.

3.2. Agile Project Management

The lecture “Communication and Strategy” in master studies Systems Engineering deepen the existing knowledge from soft skill courses and focus specifically on their application in product development projects and the management of project teams. 40 students take part each year. At the end of the course students should know the different push and pull communication styles and be able to use the right style for a specific situation. Moreover, students should know different communication media (e.g. chat, web conference, virtual reality meetings) and use them properly in different communication situations. Finally, they should be able to strategically prepare a specific project meeting, i.e. define their objectives, analyze the communication partner, choose the right media, and prepare the fitting communication style and a backup plan. According to Bloom’s taxonomy level 6 “evaluation” can be reached.

In the old concept, the learning content is acquired in a workshop setting. During three days the lecturer presents the theory and lets the student apply it in role plays in small groups. The last workshop is completely organized by the student teams in which they have to prove a given research hypothesis. So far, digital tools have been used in the workshop. HTC VIVEs are available for carrying out design reviews in the virtual reality software WeAre-rooms (cp. Fig. 4Fig.), web conferences are practiced with Adobe Connect and chats are performed with WhatsApp. However, due to corona pandemic the workshop setting could not be maintained.

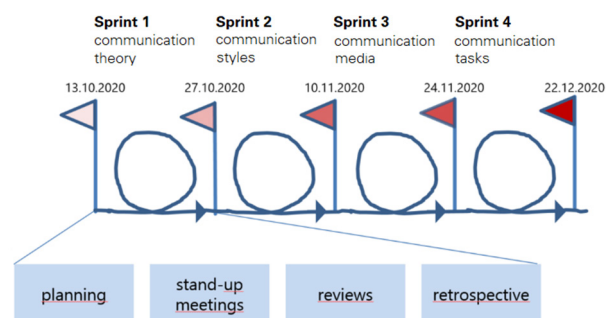


Fig. 5 Definition of eduScrum sprints.

In the new concept the lecture was organized as an eduScrum project, following [4]. Thus, the agile project management method scrum was adjusted to the circumstances of a lecture. The lecturer took the role as product owner and the student team was the scrum team. One student of the student team has been selected as scrum master. According to Fig. 5 the lecture consisted of four sprints. Each sprint was planned for two weeks and contains a meeting once a week. Each sprint starts with a kick-off meeting. The product owner presents two user stories for each group. From the user story students develop their celebration criteria and plan their individual tasks. The scrum team presents their planning and current status during the stand-up meeting in the following week. This allows the product owner to precise the user story and the team to adjust the planning. In the following week the teams present their results and the product owner states the user stories for the next sprint. From one sprint to the other student teams work more and more independently and the user stories become more and more fuzzy. In the first sprint a lot of theory is explained in videos and provided through the e-learning platform Moodle. The students need to understand (Bloom level 1). The learning complexity is successively increased. In the fourth sprint no theory is explained by the lecturer. Students need to search and create their own content, explain it to the other students and evaluate, if the others did understand.

Although, scrum is supported by the project management software as used in 3.1 and 3.3 it has been decided to use a simplified approach to keep the administration effort low. All information has been bundled in the e-learning platform Moodle. Meeting rooms are created with the BigBlueButton plug-in. The PDF-Annotation plug-in provides an easy-to-use possibility for a Kanban board. Fig. 6 shows the current status of one student team at sprint halftime. The frames are realized with an underlying PDF file. The tickets are realized through annotations (the blue bubbles). The author of an annotation is responsible for the task described within. The annotations can be dragged to another place on the PDF, i.e. moved from “to do” to “busy” to “done”.

The introduction of an agile project management method can be seen as a success. Students were forced to question their understanding of the task (user story) right at the beginning of a sprint to define celebration criteria. Thus, the lecturer did not generate a huge task description taking into account all possible misunderstandings. The clarification of the user story is part of the task. Since students continuously work on their tasks a

lecture Communication and Strategy in Engineering Projects		team name Social Network Based Design	team members	
stories	celebration criteria	to do	busy	done
  	    	  	  	  
definition of done   	definition of fun     	run up chart		impediments 

Fig. 6: Realisation of a Kanban board in Moodle.

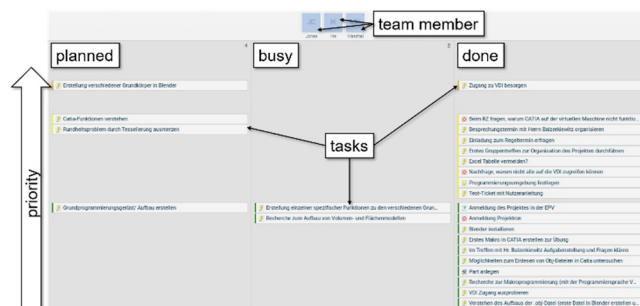


Fig. 7 Kanban-board of a student project team.

continuous communication took place and misunderstandings could be cleared up before the end of the task. The eduScrum procedure with regular communication meetings helped to keep the close contact to the students and allowed to keep the learning complexity on the intended Bloom's level 6.

In the student evaluation, the lecture has an overall rating on approximately the same level as the lecture before corona pandemic. Thus, the negative effect could be averted but there seems to be no additional positive effect. However, some students praised the approach for giving a structure and keep the effort evenly distributed throughout the semester. Especially the sprint duration of two weeks was highlighted as a good solution. However, the lecturer performed the role of a coach and students thought each other. Feedback culture and the inclusion of own ideas led to higher motivation, cp. also [8].

Working with PDF-Annotations to simulate a Kanban board was just a compromise solution. It worked, but it didn't provide a good overview and has no automation, e.g. to provide a burn down chart. Thus, for the following semester it was decided to use the scrum functions of the earlier described software Projektron BCS.

The main disadvantage was the inability to use virtual reality. Students don't have VR equipment at home, on the other hand they were not allowed to enter the university in groups. Thus, the real cooperation in VR has been restricted to a small group of students. The other students just got a feeling of VR by streaming the desktop view in a web conference.

3.3. Hybrid Project Management

During mechanical engineering studies, students need to perform several student theses. In the fifth semester of the bachelor studies two student thesis are performed and in the seventh semester the bachelor thesis needs to be written. During master studies an interdisciplinary team project is performed during two semesters and the studies are completed with the master thesis.

The tasks of a student thesis are individual and defined between the student and the lecturer. Usually, students create a project plan on the basis of the task description and report regularly to the lecturer in a personal meeting. The dates and contents are planned by the students themselves.

Due to the pandemic no unintentional meetings (e.g. before or after a regular lecture or during lunch in the cafeteria) took place. The mental hurdle to organize a video conference was very high. Sometimes students get lost by concentrating on

unimportant details or trying to solve unsolvable issues. As a result students lose time, the project gets stuck and the objective are not met. Thus, a hybrid project management approach should maintain the personal responsibility for their own thesis, but should provide a frame for regular communication and real time problem solving.

As a new concept to deal with these new challenges a hybrid project management was introduced to student projects. As a stripped-down version of classic project management, students create a milestone plan and respective tasks fitting to their individual project. The advantages of agile project management are used through tickets that are planned within the upcoming task. The tickets are organized on a Kanban-board that provides five different states:

1. New: Tickets that need to be done in the recent task, but have no due date planned yet.
2. Planned: Tickets that own a specific due date and need treatment in the next period.
3. Busy: Tickets that are treated in the recent period.
4. Done: Tickets that are closed by the students, but need to be reported to the supervisor.
5. Closed: Tickets that are discussed with supervisor and completely closed.

The tickets are reported in regular weekly meetings. The meeting is scheduled for a maximum of 30 minutes and up to ten students. Thus, each students has up to 3 minutes to report. Each student reports a short status of the tickets in done. If the supervisor agrees the ticket is moved to closed. Afterwards tickets are moved from planned to busy. That means these tickets need to be treated until the next meeting. If a ticket cannot be closed as planned it is moved to the top giving it a high priority and visualized by a red badge. If the student is not able to solve the issue himself the supervisor gets a ticket to support the solution finding process. Since the specific solution is usually not relevant to the complete team, the discussion is done directly after the meeting in a smaller group during a table discussion. Fig. 7 shows a snapshot of a student's Kanban board at the beginning of the regular meeting.

The tasks and tickets are generated in the project management software Projektron BCS. Each student owns a project as project leader. The supervisor is able to view the student projects. Since students work steadily in their projects the supervisor is able to see the progress in real-time. Fig. 8Fig. shows the ticket overview one day before the regular meeting. It can be recognized that two tickets still need treatment

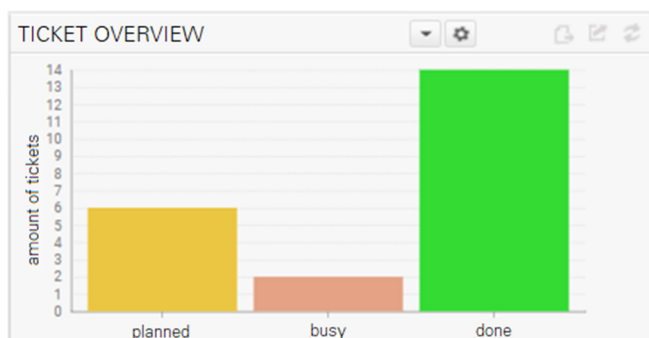


Fig. 8 Student team ticket overview showing planned, busy and done tasks

whereas 14 tickets are already done. The project seems to be on time.

The weekly meetings are performed in Adobe Connect. Each student shares the screen and presents the Kanban-board. The role of the supervisor is just to moderate the meeting and ask questions whenever a task is not clear or a ticket haven't reached the definitions of done.

In order to evaluate the use of the hybrid project management for student thesis, students were asked to agree to predefined statements (multiple answers allowed). All respondents agreed to these statements:

- It helped to structure my own work.
- It gave me a good feeling on being on the right track.
- It helped to get immediate support, when needed.

A majority also agreed to these statements:

- It helped to get new ideas through exchange with others.
- It was a good alternative during the pandemic.

Just one respondent thinks the regular meeting is a time waster. Nobody agreed to the following statements:

- It did not help me at all.
- It helped to schedule my time.

Overall, 71.4% of the students rate the method as very good and 28.5% of the students rate the method as good. Therefore, the hybrid project management approach can be seen as a suitable method to organize student theses also without any presence contact. The method enables the students to pursue their own topics and to realize ideas and at the same time to be integrated into a community and receive support if necessary. According to [9] this approach increases intrinsic motivation and thus leads to better results.

One difficulty is due to the lacking support and integration of video conferencing directly into the project management software. The conference had to be created in Adobe Connect and participants had to share their screens. Since each student had a scheduled presentation time of just 3 minutes, waiting for the screen to be visible every 3 minutes is quite annoying. Furthermore, there is no connection between participants of the web meeting and in the project management software. Thus, agenda, moderation, administration etc. needs to be performed by permanently switching between to different applications.

4. Conclusion and Outlook

The Corona pandemic forced universities to transits attendance lectures into e-learning formats from one day to the other. It was shown in this contribution that lectures benefit from both, classic and agile, project management approaches. Even student theses profit from the here shown hybrid project management approaches. However, what was shown here is not the result from a carefully planned implementation project. Although the results are promising no systematic evaluation was done. The case study allows to state following problems:

- The use of the tools must be learned. The time required for this is no longer available to work on the actual learning content.
- There is no single platform that combines all the necessary functionalities. The administration of the various solutions costs time and money.
- Students have different access to digital tools. It is difficult to address diversity in a distributed teaching.

Therefore, a didactic concept must be developed that enables students to use the needed functionalities without causing frustration. The teachers also need support in using the tools. Tool-integrated assistance systems can support teachers in their communication tasks. In addition, the technology must be accessible to all students. Since VR headsets cannot be made available to all students, options for using the technology with less immersion (e.g. on the screen or with the help of smartphone and cardboard) must be checked. For reasons of sustainable implementation, it must also be ensured that license fees can be financed on a permanent basis and administration personal can be paid.

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